



Chapter 6 Cluster Analysis



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§ 6.1 Introduction

- Discriminant analysis and cluster analysis are two classification methods with different purposes and functions.
- In discriminant analysis, the number of groups is **known**, and we allocate samples to one of the pre-defined groups (or classes). However, in cluster analysis, both the number of clusters and the clusters themselves are typically **unknown** beforehand.
- Cluster Analysis: Divide objectives into a number of groups. Similar objectives are in the same group and the dissimilar ones are in different groups.
- Cluster analysis has two types: **Q Type** (classification objectives are samples) and **R Type** (classification objectives are variables).

§ 6.2 Distance and Similarity Coefficient

- ▶ Similarity Measurement: Distance and Similarity Coefficient.
- ▶ Distance is usually used to measure the similarity of samples and similarity coefficient is used to measure the similarity of variables.
- ▶ There exist many different definitions of distance and similarity coefficient. These definitions have close relationship with the types of variables.

Measurement Scale of Variables

- ➡ Variables can be classified to three types, interval, orderly and nominal variables.
- **Interval variables (间隔变量)**: variables are denoted by continuous quantities, like length, weight, speed and temperature.
- **Orderly Scale variables (有序变量)**: variables are measured not by specific quantity but by grades. For instance, a certain product is divided into first, second and third quality goods.
- **Nominal Scale variables (名义变量)**: variables are divided into clusters. There exist no grade relation and no quantitative relation, like genders, occupations, types of product.
- This chapter mainly discusses sample cluster analysis method with interval variable.

§ 6.3 Hierarchical Cluster (系统聚类)Method

- **Hierarchical Clustering Method** is the most used method among all the cluster analysis methods.
- Basic idea:
 - Step 1: treat each of n sample as a cluster and stipulate the sample-to-sample and cluster-to-cluster distances.
 - Step 2: combine the two clusters with the smallest distance into a new cluster and calculate the distance between the new cluster and others respectively ;
 - Step 3: reduce a cluster each time by repeating step 2 until all the sample are in one cluster.
- several kinds of methods will be introduced. The difference lies in the way the distance between clusters is calculated

6 methods for cluster analysis

- 1、Shortest distance method
- 2、Longest distance method
- 3、Average linkage (类平均)method
- 4、Centroid (重心) method
- 5、Sum of squares of deviations (离差平方和) method (Ward method)
- 6、Unity of Hierarchical Clustering Method
- 7、Numbers of clusters

1. Shortest distance method

- Define the distance between two clusters as the shortest distance between samples of the two clusters:

$$D_{KL} = \min_{i \in G_K, j \in G_L} d_{ij}$$



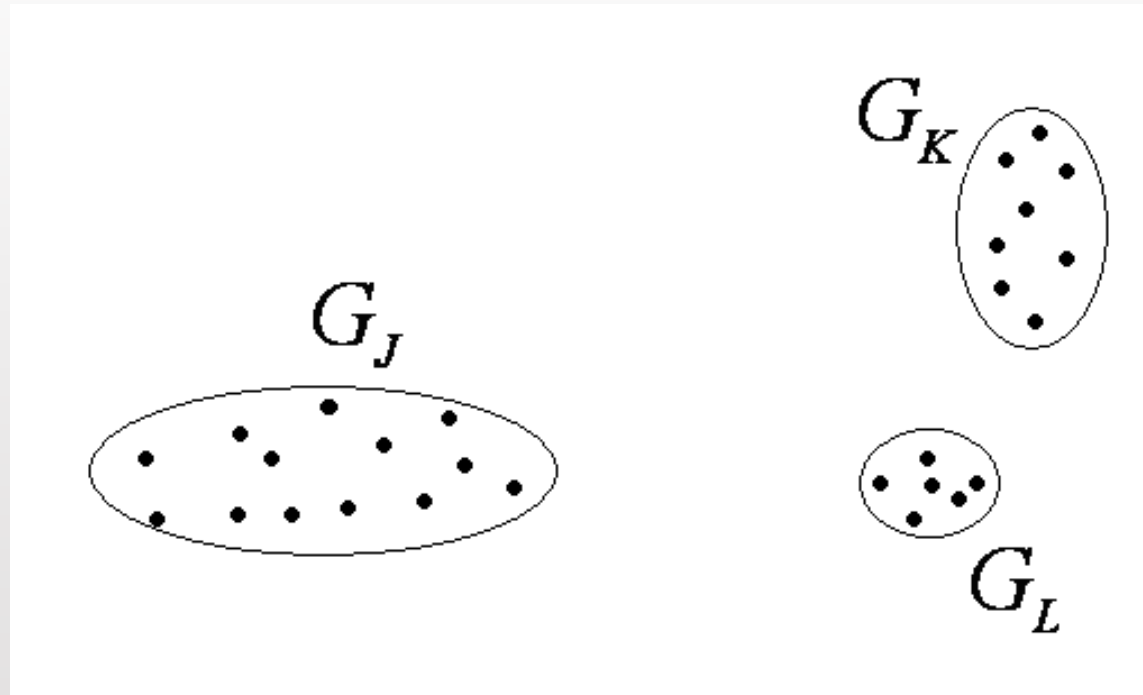
Figure 6.3.1 Shortest distance method: $D_{KL} = d_{23}$

Clustering Procedure of shortest distance method

- ➡ (1) Define the distance between samples. Calculate the distance matrix of n samples, $\mathbf{D}_{(0)}$, which is a symmetric matrix.
- ➡ (2) Choose the smallest element of $\mathbf{D}_{(0)}$, D_{KL} . Combine G_K and G_L into a new cluster, G_M . Then, $G_M = G_K \cup G_L$.
- ➡ (3) The recursion formula to calculate the distances between G_M and any other cluster G_J is

$$\begin{aligned} D_{MJ} &= \min_{i \in G_M, j \in G_J} d_{ij} = \min \left\{ \min_{i \in G_K, j \in G_J} d_{ij}, \min_{i \in G_L, j \in G_J} d_{ij} \right\} \\ &= \min \{ D_{KJ}, D_{LJ} \} \end{aligned} \quad (6.3.2)$$

Graphical Representation of Recursion Formula

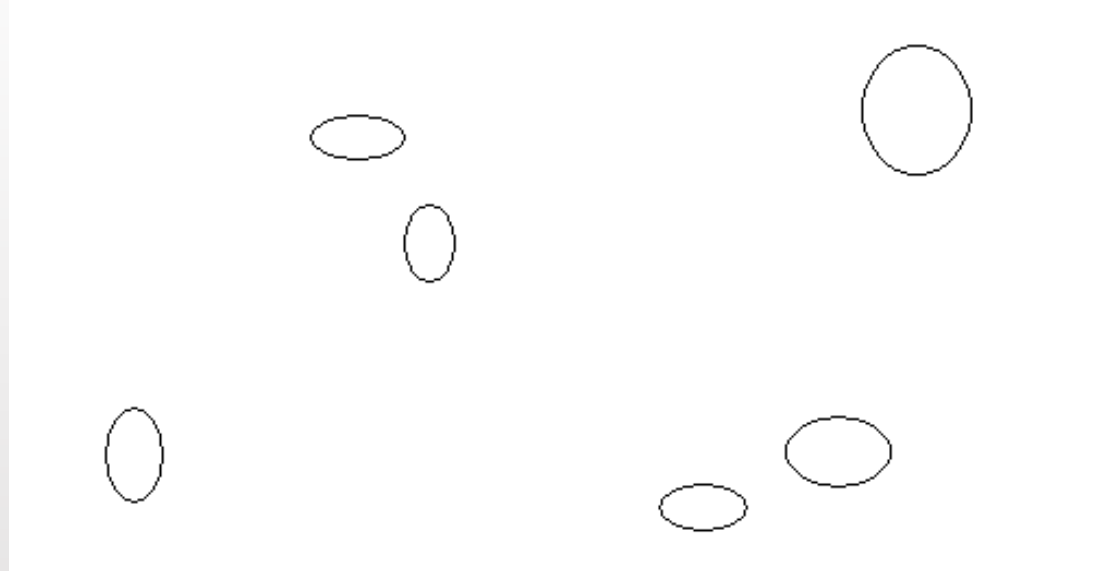


Clustering Procedure of Shortest Distance Method

In $\mathbf{D}_{(0)}$, the columns and rows of G_K and G_L are respectively combined into a new column and row, G_M . The new distance value between G_M and any other cluster G_J can be obtained from (6.3.2). The distance values from other rows and columns remain unchanged. Then, we can get the new distance matrix, $\mathbf{D}_{(1)}$.

- ➡ (4) Repeat the above two steps to obtain $\mathbf{D}_{(2)}$, and so on, until all the elements are in one cluster.

- If more than one smallest elements $D_{(m)}$ exists, the situation is called **Tie**.
(结) Any couple or all of the clusters with the smallest elements can be combined into one cluster. Tie usually happens in shortest distance method.



- Since shortest distance method is based on the shortest distance between the samples of the two clusters. Thus, this method is not applicable to clusters with poor separation.

- ➡ **Example 6.3.1** Suppose indexes of 5 samples are respectively 1, 2, 6, 8, 11. Please use shortest distance method to cluster them.
- Denote $G_1=\{1\}$, $G_2=\{2\}$, $G_3=\{6\}$, $G_4=\{8\}$, $G_5=\{11\}$. Use absolute distance between samples.

Table 6.3.1

	$D_{(0)}$				
	G_1	G_2	G_3	G_4	G_5
G_1	0				
G_2	1	0			
G_3	5	4	0		
G_4	7	6	2	0	
G_5	10	9	5	3	0

Table 6.3.2

 $D_{(1)}$

	G_6	G_3	G_4	G_5
G_6	0			
G_3	4	0		
G_4	6	2	0	
G_5	9	5	3	0

where $G_6 = G_1 \cup G_2 = \{1, 2\}$.

Table 6.3.3

 $D_{(2)}$

	G_6	G_7	G_5
G_6	0		
G_7	4	0	
G_5	9	3	0

where $G_7 = G_3 \cup G_4 = \{6, 8\}$.

Table 6.3.4

 $D_{(3)}$

	G_6	G_8
G_6	0	
G_8	4	0

where $G_8 = G_5 \cup G_7 = \{6, 8, 11\}$.

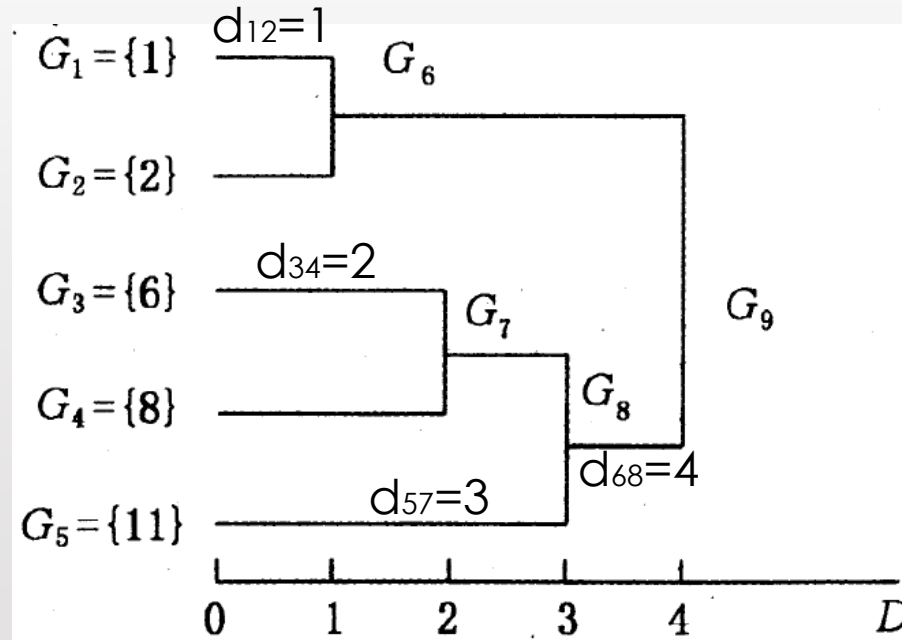


Figure 6.3.2 Tree Diagram

From this diagram, it is suitable to divide all samples into two cluster

2. Longest Distance Method

- Define the distance between two clusters as the longest distance between samples of the two clusters.

$$D_{KL} = \max_{i \in G_K, j \in G_L} d_{ij}$$

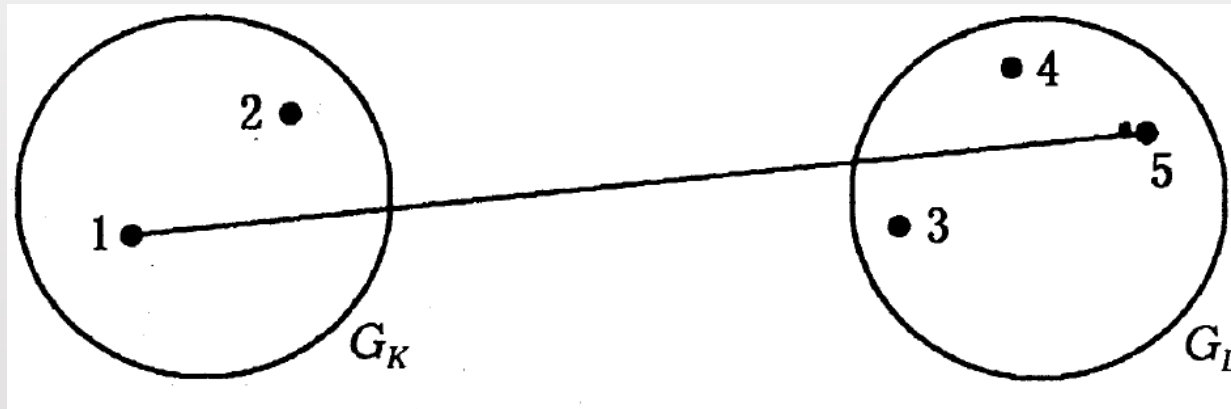
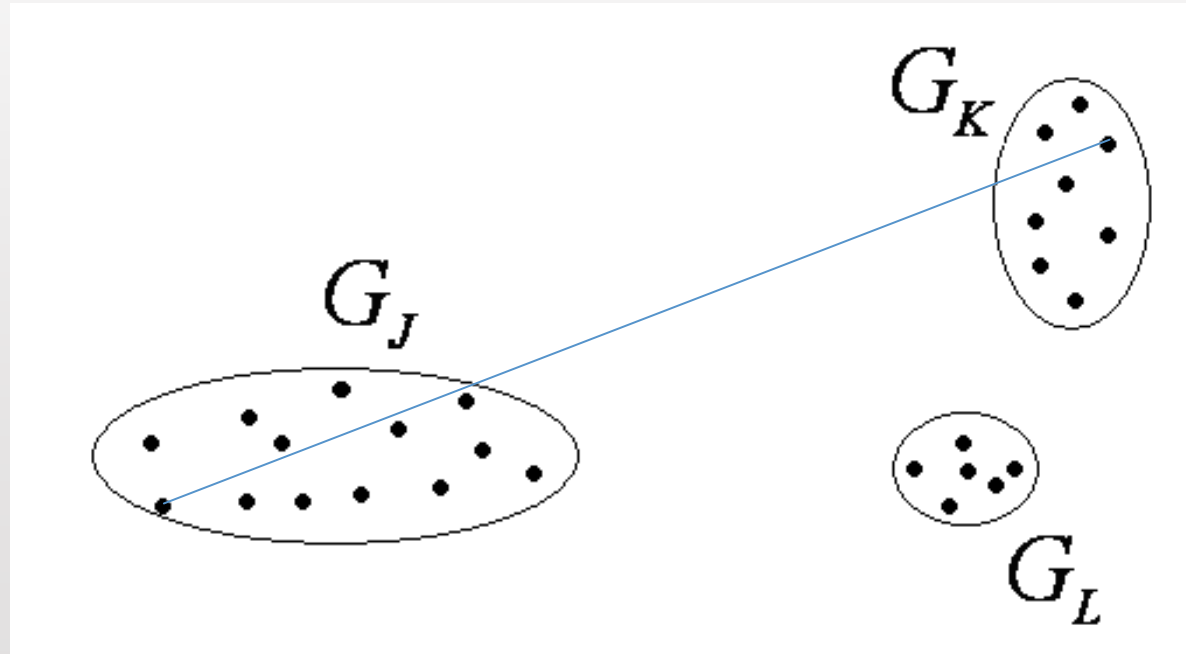


Figure 6.3.3 Longest Distance Method: $D_{KL} = d_{15}$

- The procedure of longest distance method is exactly the same as shortest distance method. The only difference is recursion formula of cluster distance.

- Recursion Formula :

$$D_{MJ} = \max \{ D_{KJ}, D_{LJ} \}$$



Exercise:

- ▶ 6.1 For example 6.3.1, please using the longest distance method to perform cluster analysis and give the tree diagram for cluster analysis.

Denote by $G_1=\{1\}$, $G_2=\{2\}$, $G_3=\{6\}$, $G_4=\{8\}$, $G_5=\{11\}$.

Use absolute distance between samples.

- Use longest distance method on Example 6.3.1 and its tree diagram as Figure 6.3.4. It has similar form as Figure 6.3.2, but its distances used to combine clusters are longer than that in Figure 6.3.2. It is better to divide samples into two clusters from the figure.

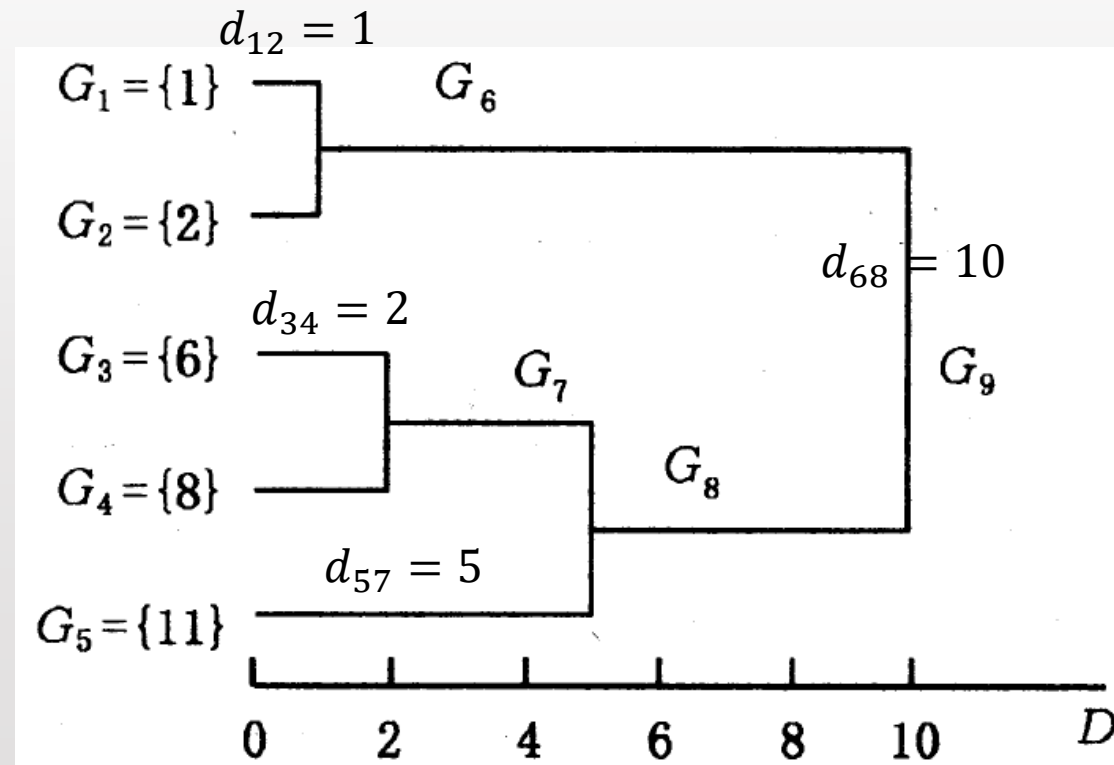
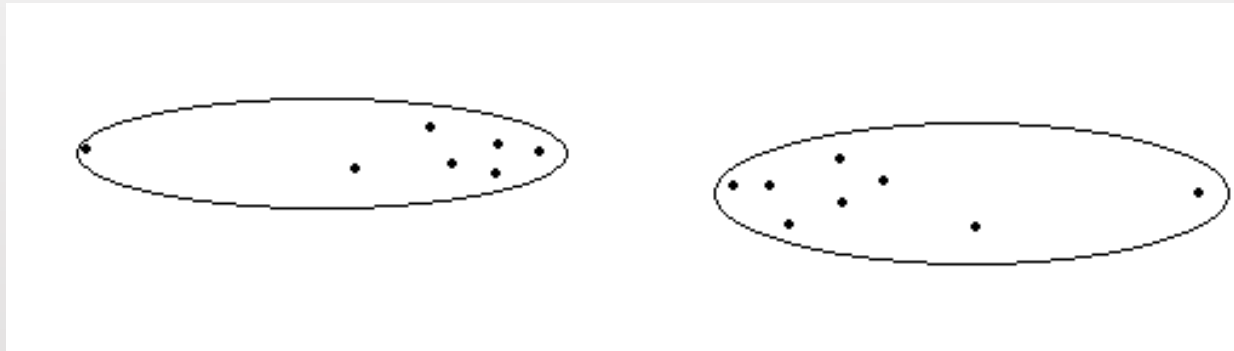


Figure 6.3.4 Tree Diagram

Influence of Abnormal Values

- ▶ Longest distance method can be seriously twisted by abnormal values. A effective method is to separate out the abnormal values first and then cluster the samples.



3. Average Linkage(类平均)Method

- ➡ **Average Linkage Method** has two definitions. One is using average distance of all samples between two clusters as the distance between clusters. Then the distance between G_K and G_L

$$D_{KL} = \frac{1}{n_K n_L} \sum_{i \in G_K, j \in G_L} d_{ij} \quad (6.3.5)$$

where n_K and n_L are respectively the sample numbers of G_K and G_L , d_{ij} is the distance between the i -th sample from G_K and the k -th G_L , shown as Figure 6.3.7. Its recursion formula is:

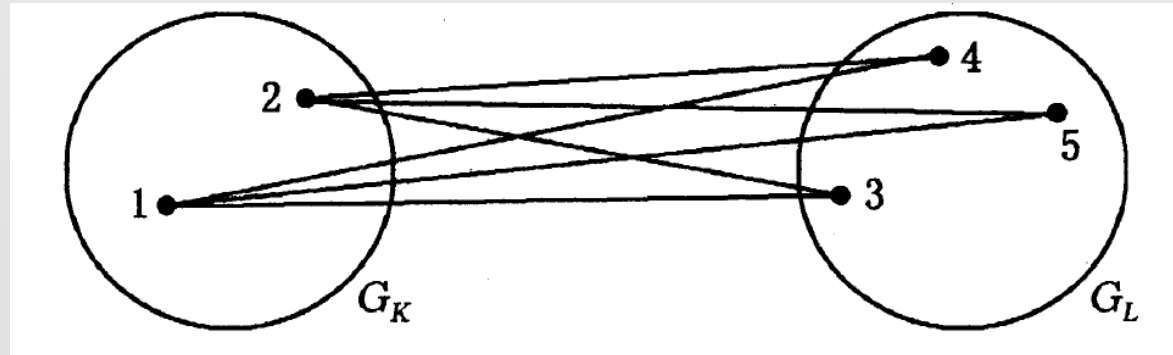


Figure 6.3.7 $D_{KL} = (d_{13} + d_{14} + d_{15} + d_{23} + d_{24} + d_{25}) / 6$

Let D_M be the new cluster obtained by combining D_k and D_l .

We have the following recursion formula :

$$\begin{aligned} D_{MJ} &= \frac{1}{n_M n_J} \sum_{i \in G_M, j \in G_J} d_{ij} = \frac{1}{n_M n_J} \left(\sum_{i \in G_K, j \in G_J} d_{ij} + \sum_{i \in G_L, j \in G_J} d_{ij} \right) \\ &= \frac{n_K}{n_M} D_{KJ} + \frac{n_L}{n_M} D_{LJ} \end{aligned} \tag{6.3.6}$$

- The other method defines the square distance between clusters as the average square distance between samples from the two clusters:

$$D_{KL}^2 = \frac{1}{n_K n_L} \sum_{i \in G_K, j \in G_L} d_{ij}^2 \quad (6.3.7)$$

- Its recursion formula is

$$D_{MJ}^2 = \frac{n_K}{n_M} D_{KJ}^2 + \frac{n_L}{n_M} D_{LJ}^2 \quad (6.3.8)$$

- Average linkage method take the advantage of information of all the samples. In many situations, it is known as a better hierarchical clustering method.

- Use (squared) average linkage method in (6.3.7) on Example 6.3.1 to cluster. In the beginning, square every element of $\mathbf{D}_{(0)}$ as $\mathbf{D}_{(0)}^2$.

Table 6.3.6

$\mathbf{D}_{(0)}^2$

	G_1	G_2	G_3	G_4	G_5
G_1	0				
G_2	1	0			
G_3	25	16	0		
G_4	49	36	4	0	
G_5	100	81	25	9	0

Denote by $G_1=\{1\}$, $G_2=\{2\}$, $G_3=\{6\}$, $G_4=\{8\}$, $G_5=\{11\}$.

Table 6.3.7

$D_{(1)}^2$

	G_6	G_3	G_4	G_5
G_6	0			
G_3	20.5	0		
G_4	42.5	4	0	
G_5	90.5	25	9	0

$$D_{63}^2 = \frac{1}{2} D_{13}^2 + \frac{1}{2} D_{23}^2 = \frac{1}{2} (5^2 + 4^2) = 20.5, \quad D_{64}^2 = \frac{1}{2} D_{14}^2 + \frac{1}{2} D_{24}^2 = \frac{1}{2} (7^2 + 6^2) = 42.5$$

Table 6.3.8

$D_{(2)}^2$

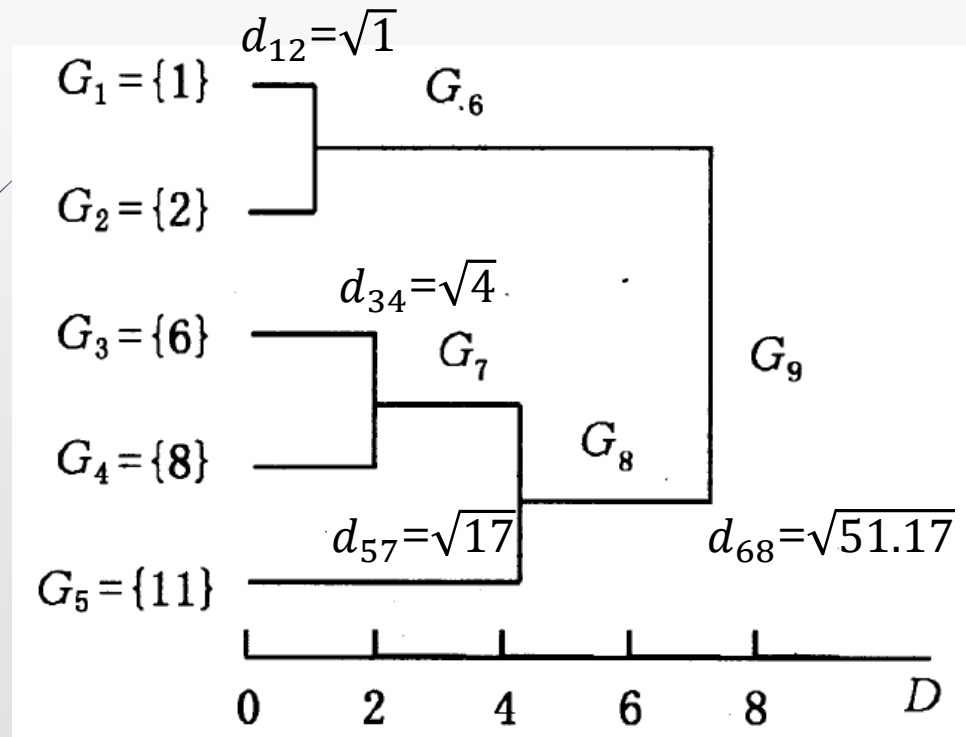
	G_6	G_7	G_5
G_6	0		
G_7	31.5	0	
G_5	90.5	17	0

where $G_7 = G_3 \cup G_4 = \{6, 8\}$. $D_{76}^2 = \frac{1}{4} (5^2 + 4^2 + 7^2 + 6^2) = 31.5$

Table 6.3.9

 $D_{(3)}^2$

	G_6	G_8
G_6	0	
G_8	51.17	0



$$D_{86}^2 = \frac{1}{6} (5^2 + 4^2 + 7^2 + 6^2 + 10^2 + 9^2) = 51.17,$$

Figure 6.3.8 Tree Diagram

Assignment:

- ▶ 6.1 For example 6.3.1, please use the class average method (6.3.5) to perform cluster analysis and give the tree diagram for cluster analysis.

Denote by $G_1=\{1\}$, $G_2=\{2\}$, $G_3=\{6\}$, $G_4=\{8\}$, $G_5=\{11\}$.
The distance between G_K and G_L is given by

$$D_{KL} = \frac{1}{n_K n_L} \sum_{i \in G_K, j \in G_L} d_{ij} \quad (6.3.5)$$

4. Centroid (重心) Method

- Define the distance between two clusters as the Euclidean distance between the centers of gravity of the two clusters. Suppose the centers of gravity of G_K and G_L are respectively $\bar{\mathbf{x}}_K$ and $\bar{\mathbf{x}}_L$, then the squared distance between G_K and G_L is

$$D_{KL}^2 = d_{\bar{\mathbf{x}}_K \bar{\mathbf{x}}_L}^2 = (\bar{\mathbf{x}}_K - \bar{\mathbf{x}}_L)' (\bar{\mathbf{x}}_K - \bar{\mathbf{x}}_L)$$

- The center of gravity of the new cluster G_M after merging G_K and G_L is

$$\bar{\mathbf{x}}_M = \frac{n_K \bar{\mathbf{x}}_K + n_L \bar{\mathbf{x}}_L}{n_M}, \quad (6.3.10)$$

where $n_M = n_K + n_L$ is the number of samples in G_M

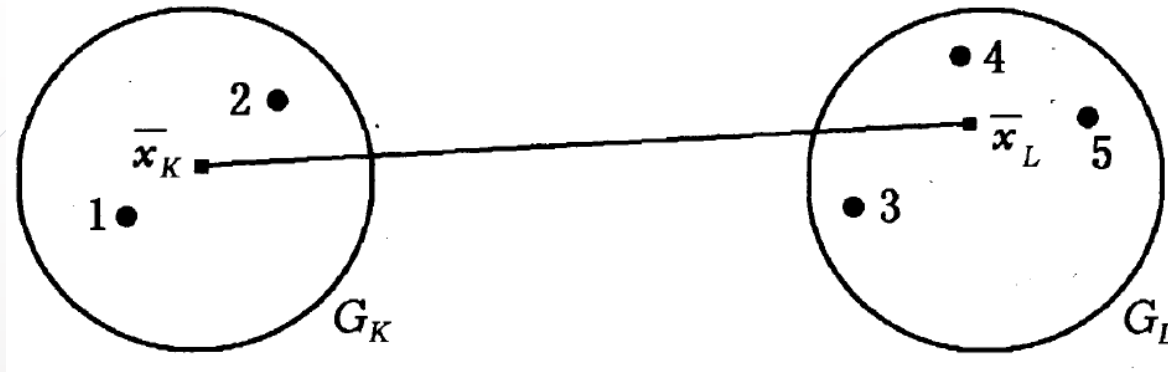


Figure 6.3.9 Centroid Method: $D_{KL} = d_{\bar{x}_K \bar{x}_L}$

- Compared to other hierarchical clustering methods, centroid method is more stable on handling abnormal values. But, it is not comparable to average linkage method or Ward method in other aspects.

4. Centroid (重心) Method

- ▶ The recursion formula of centroid method (Appendix 6-2, I) is

$$D_{MJ}^2 = \frac{n_K}{n_M} D_{KJ}^2 + \frac{n_L}{n_M} D_{LJ}^2 - \frac{n_K n_L}{n_M^2} D_{KL}^2 \quad (6.3.11)$$

一、(6.3.11)式的推导

设某一步将 G_K 和 G_L 合并成 G_M , 它们的重心分别是 $\bar{x}_K, \bar{x}_L$ 和 $\bar{x}_M$, 它们各有 n_K, n_L 和 n_M ($n_M = n_K + n_L$) 个样品, 显然

$$\bar{x}_M = \frac{1}{n_M} (n_K \bar{x}_K + n_L \bar{x}_L)$$

新类 G_M 与任一类 G_J (重心为 $\bar{x}_J$) 的平方距离是

$$\begin{aligned} D_{MJ}^2 &= (\bar{x}_J - \bar{x}_M)' (\bar{x}_J - \bar{x}_M) \\ &= \left[\bar{x}_J - \frac{1}{n_M} (n_K \bar{x}_K + n_L \bar{x}_L) \right]' \left[\bar{x}_J - \frac{1}{n_M} (n_K \bar{x}_K + n_L \bar{x}_L) \right] \\ &= \bar{x}_J' \bar{x}_J - 2 \cdot \frac{n_K}{n_M} \bar{x}_J' \bar{x}_K - 2 \cdot \frac{n_L}{n_M} \bar{x}_J' \bar{x}_L + \frac{1}{n_M^2} (n_K^2 \bar{x}_K' \bar{x}_K + 2n_K n_L \bar{x}_K' \bar{x}_L + n_L^2 \bar{x}_L' \bar{x}_L) \\ &= \frac{n_K}{n_M} (\bar{x}_J' \bar{x}_J - 2\bar{x}_J' \bar{x}_K + \bar{x}_K' \bar{x}_K) + \frac{n_L}{n_M} (\bar{x}_J' \bar{x}_J - 2\bar{x}_J' \bar{x}_L + \bar{x}_L' \bar{x}_L) \end{aligned}$$

4. Centroid (重心) Method

- The recursion formula of centroid method (Appendix 6-2, I) is

$$D_{MJ}^2 = \frac{n_K}{n_M} D_{KJ}^2 + \frac{n_L}{n_M} D_{LJ}^2 - \frac{n_K n_L}{n_M^2} D_{KL}^2 \quad (6.3.11)$$

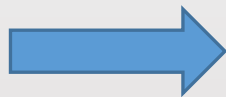
(Continue)

$$\begin{aligned} & -\frac{n_K n_L}{n_M^2} (\bar{\mathbf{x}}_K' \bar{\mathbf{x}}_K - 2\bar{\mathbf{x}}_K' \bar{\mathbf{x}}_L + \bar{\mathbf{x}}_L' \bar{\mathbf{x}}_L) \\ &= \frac{n_K}{n_M} (\bar{\mathbf{x}}_J - \bar{\mathbf{x}}_K)' (\bar{\mathbf{x}}_J - \bar{\mathbf{x}}_K) + \frac{n_L}{n_M} (\bar{\mathbf{x}}_J - \bar{\mathbf{x}}_L)' (\bar{\mathbf{x}}_J - \bar{\mathbf{x}}_L) \\ & \quad - \frac{n_K n_L}{n_M^2} (\bar{\mathbf{x}}_K - \bar{\mathbf{x}}_L)' (\bar{\mathbf{x}}_K - \bar{\mathbf{x}}_L) \\ &= \frac{n_K}{n_M} D_{KJ}^2 + \frac{n_L}{n_M} D_{LJ}^2 - \frac{n_K n_L}{n_M^2} D_{KL}^2 \end{aligned}$$

5. Sum of square of deviations method (Ward method)

- The sum of square Euclidean distances **between all the samples and the cluster's center of gravity** is called sum of square of deviations method(离差平方和) within cluster. Combine G_K and G_L into new cluster G_M . Then, sum of square of deviations of G_K , G_L and G_M are respectively

离差平方和:类中各样品到类重心的平方欧氏距离之和



$$W_K = \sum_{i \in G_K} (\mathbf{x}_i - \bar{\mathbf{x}}_K)' (\mathbf{x}_i - \bar{\mathbf{x}}_K)$$

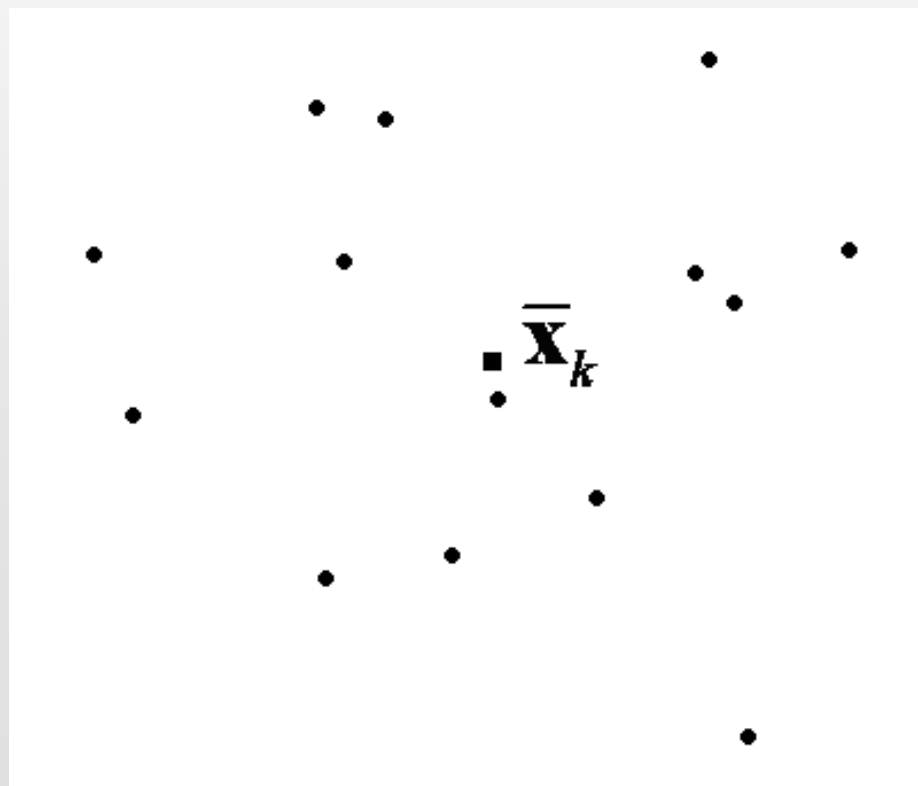
$$W_L = \sum_{i \in G_L} (\mathbf{x}_i - \bar{\mathbf{x}}_L)' (\mathbf{x}_i - \bar{\mathbf{x}}_L)$$

$$W_M = \sum_{i \in G_M} (\mathbf{x}_i - \bar{\mathbf{x}}_M)' (\mathbf{x}_i - \bar{\mathbf{x}}_M)$$

They reflect degree of dispersion of samples of clusters.

Geometric Explanation of Sum of Square of Deviation of Inside Cluster

- Sum of square of deviation of inside cluster W_K is sum of square of linear distances between all the samples in G_K and center of gravity $\bar{\mathbf{x}}_k$.



- The distances between the two classes, G_K and G_L , are close, then the increase in the sum of squared deviations after merging should be relatively small; otherwise, it should be larger. Thus, we define the square distance between G_K and G_L as

$$D_{KL}^2 = W_M - W_K - W_L \quad (6.3.14)$$

This kind of hierarchical clustering method (系统聚类法) is called as **Sum of Square of Deviation Method**(离差平方和法) or **Ward's minimum variance method** (Ward方法).

- D_{KL}^2 can also be shown as

$$D_{KL}^2 = \frac{n_K n_L}{n_M} (\bar{\mathbf{x}}_K - \bar{\mathbf{x}}_L)' (\bar{\mathbf{x}}_K - \bar{\mathbf{x}}_L) \quad (6.3.15)$$

Proof:

$$\begin{aligned} W_M &= \sum_{i \in GM} (\mathbf{x}_i - \bar{\mathbf{x}}_M)' (\mathbf{x}_i - \bar{\mathbf{x}}_M) \\ &= \sum_{i \in GM} (\mathbf{x}_i - \bar{\mathbf{x}}_K + \bar{\mathbf{x}}_K - \bar{\mathbf{x}}_M)' (\mathbf{x}_i - \bar{\mathbf{x}}_K + \bar{\mathbf{x}}_K - \bar{\mathbf{x}}_M) \\ &= W_K + W_L + \frac{n_L n_K}{n_M} (\bar{\mathbf{x}}_K - \bar{\mathbf{x}}_L)' (\bar{\mathbf{x}}_K - \bar{\mathbf{x}}_L) \end{aligned}$$

$$\bar{\mathbf{x}}_M = \frac{n_K \bar{\mathbf{x}}_K + n_L \bar{\mathbf{x}}_L}{n_M}$$



Remarks: In cluster analysis, the sum of squares of deviation method (离差平方和法) often results in larger distances between two significant clusters, making their combination challenging. Conversely, smaller clusters typically exhibit shorter distances, facilitating their merger. This pattern aligns well with our practical needs in cluster analysis.



Denote by $G_1=\{1\}$, $G_2=\{2\}$, $G_3=\{6\}$, $G_4=\{8\}$, $G_5=\{11\}$.

- The following square distance recursion formula of sum of squares of deviations method can be derived from (6.3.15) and (6.3.11) (Appendix 6-2, III):

$$D_{MJ}^2 = \frac{n_J + n_K}{n_J + n_M} D_{KJ}^2 + \frac{n_J + n_L}{n_J + n_M} D_{LJ}^2 - \frac{n_J}{n_J + n_M} D_{KL}^2 \quad (6.3.16)$$

- Apply sum of squares of deviations method to example 6.3.1:

Table 6.3.10

$D_{(0)}^2$

	G_1	G_2	G_3	G_4	G_5
G_1	0				
G_2	0.5	0			
G_3	12.5	8	0		
G_4	24.5	18	2	0	
G_5	50	40.5	12.5	4.5	0

$$D_{KL}^2 = \frac{1}{n_K n_L} \sum_{i \in G_K, j \in G_L} d_{ij}^2 \quad (6.3.7)$$

$$D_{KL}^2 = W_M - W_K - W_L \quad (6.3.14)$$

$$(1 - \frac{3}{2})^2 + (2 - \frac{3}{2})^2 = 0.5$$

$$(1 - \frac{9}{2})^2 + (8 - \frac{9}{2})^2 = 24.5$$

$$(2 - \frac{13}{2})^2 + (11 - \frac{13}{2})^2 = 40.5$$

$$(1-3)^2 + (2-3)^2 + (6-3)^2 - \left\{ \left(1-\frac{3}{2}\right)^2 + \left(2-\frac{3}{2}\right)^2 + 0 \right\} = 13.5 \text{ by (6.3.14) } D_{KL}^2 = W_M - W_K - W_L$$

Table 6.3.11

	G_6	G_3	G_4	G_5
G_6	0			
G_3	13.5	0		
G_4	28.17	2	0	
G_5	60.17	12.5	4.5	0

Denote by $G_1=\{1\}$, $G_2=\{2\}$, $G_3=\{6\}$, $G_4=\{8\}$, $G_5=\{11\}$.

➤ $G_6 = G_1 \cup G_2 = \{1, 2\}$

Table 6.3.12

	G_6	G_7	G_5
G_6	0		
G_7	30.25	0	
G_5	60.17	10.67	0

➤ $G_7 = G_3 \cup G_4 = \{6, 8\}$

$$\left(1 - \frac{17}{4}\right)^2 + \left(2 - \frac{17}{4}\right)^2 + \left(6 - \frac{17}{4}\right)^2 + \left(8 - \frac{17}{4}\right)^2 - \left\{ \left(1 - \frac{3}{2}\right)^2 + \left(2 - \frac{3}{2}\right)^2 + (6-7)^2 + (8-7)^2 \right\} = 30.25 \text{ by (6.3.14)}$$

Denote by $G_1=\{1\}$, $G_2=\{2\}$, $G_3=\{6\}$, $G_4=\{8\}$, $G_5=\{11\}$.

Table 6.3.13

		$D_{(3)}^2$	
		G_6	G_8
G_6		0	
G_8		56.03	0

➤ $G_8 = G_5 \cup G_7 = \{6, 8, 11\}$

$$(1 - \frac{28}{5})^2 + (2 - \frac{28}{5})^2 + (6 - \frac{28}{5})^2 + (8 - \frac{28}{5})^2 + (11 - \frac{28}{5})^2 - \{(1 - \frac{3}{2})^2 + (2 - \frac{3}{2})^2 + (6 - \frac{25}{3})^2 + (8 - \frac{25}{3})^2 + (11 - \frac{25}{3})^2\} = 56.03$$

by (6.3.14)

$$D_{KL}^2 = W_M - W_K - W_L$$

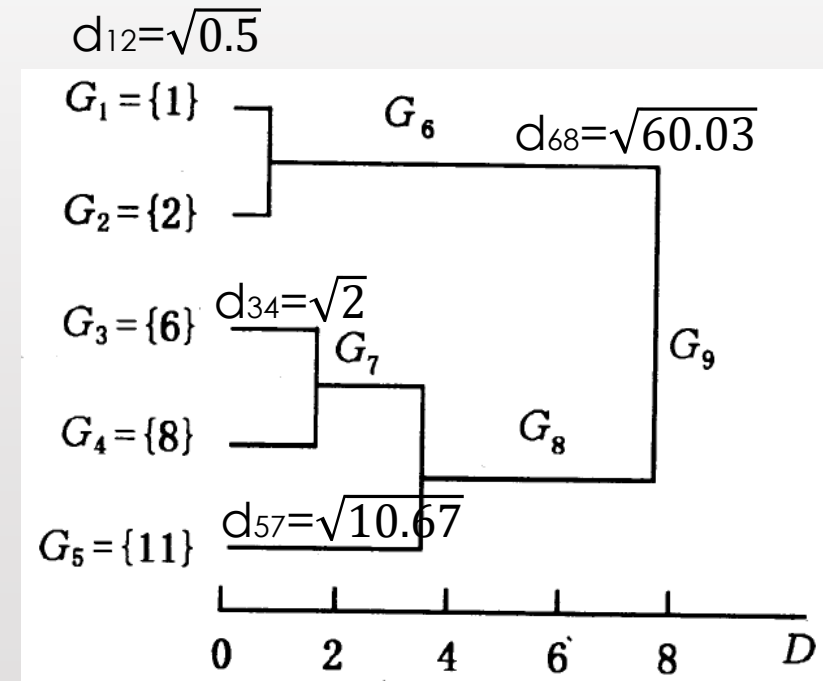


Figure 6.3.11 Tree Diagram

Example 6.3.3 In Table 6.3.14, eight main variables data of annual consumer expenditure per capita (人均年消费支出) of urban households (城镇居民家庭) from 31 provinces of year 1999 are listed. These variables are respectively:

x_1 : food x_5 : transportation and communication

x_2 : clothing x_6 : entertainment, education and cultural service

x_3 : family equipment and service x_7 : settlement

x_4 : health care x_8 : sundry (杂项) commodity and service

- Before conducting cluster analysis on data from each area, we standardize all variables, then use the shortest distance method, centroid method, and Ward method separately to conduct the cluster analysis.

Table 6.3.14

Consumer Spending Data

Unit: yuan

Area	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
北京	2959.19	730.79	749.41	513.34	467.87	1141.82	478.42	457.64
天津	2459.77	495.47	697.33	302.87	284.19	735.97	570.84	305.08
河北	1495.63	515.9	362.37	285.32	272.95	540.58	364.91	188.63
山西	1406.33	477.77	290.15	208.57	201.5	414.72	281.84	212.1
内蒙古	1303.97	524.29	254.83	192.17	249.81	463.09	287.87	192.96
辽宁	1730.84	553.9	246.91	279.81	239.18	445.2	330.24	163.86
吉林	1561.86	492.42	200.49	218.36	220.69	459.62	360.48	147.76
黑龙江	1410.11	510.71	211.88	277.11	224.65	376.82	317.61	152.85
上海	3712.31	550.74	893.37	346.93	527	1034.98	720.33	462.03
江苏	2207.58	449.37	572.4	211.92	302.09	585.23	429.77	252.54
浙江	2629.16	557.32	689.73	435.69	514.66	795.87	575.76	323.36
安徽	1844.78	430.29	271.28	126.33	250.56	513.18	314	151.39
福建	2709.46	428.11	334.12	160.77	405.14	461.67	535.13	232.29
江西	1563.78	303.65	233.81	107.9	209.7	393.99	509.39	160.12
山东	1675.75	613.32	550.71	219.79	272.59	599.43	371.62	211.84



河南	1427.65	431.79	288.55	208.14	217	337.76	421.31	165.32
湖北	1783.43	511.88	282.84	201.01	237.6	617.74	523.52	182.52
湖南	1942.23	512.27	401.39	206.06	321.29	697.22	492.6	226.45
广东	3055.17	353.23	564.56	356.27	811.88	873.06	1082.82	420.81
广西	2033.87	300.82	338.65	157.78	329.06	621.74	587.02	218.27
海南	2057.86	186.44	202.72	171.79	329.65	477.17	312.93	279.19
重庆	2303.29	589.99	516.21	236.55	403.92	730.05	438.41	225.8
四川	1974.28	507.76	344.79	203.21	240.24	575.1	430.36	223.46
贵州	1673.82	437.75	461.61	153.32	254.66	445.59	346.11	191.48
云南	2194.25	537.01	369.07	249.54	290.84	561.91	407.7	330.95
西藏	2646.61	839.7	204.44	209.11	379.3	371.04	269.59	389.33
陕西	1472.95	390.89	447.95	259.51	230.61	490.9	469.1	191.34
甘肃	1525.57	472.98	328.9	219.86	206.65	449.69	249.66	228.19
青海	1654.69	437.77	258.78	303	244.93	479.53	288.56	236.51
宁夏	1375.46	480.89	273.84	317.32	251.08	424.75	228.73	195.93
新疆	1608.82	536.05	432.46	235.82	250.28	541.3	344.85	214.4

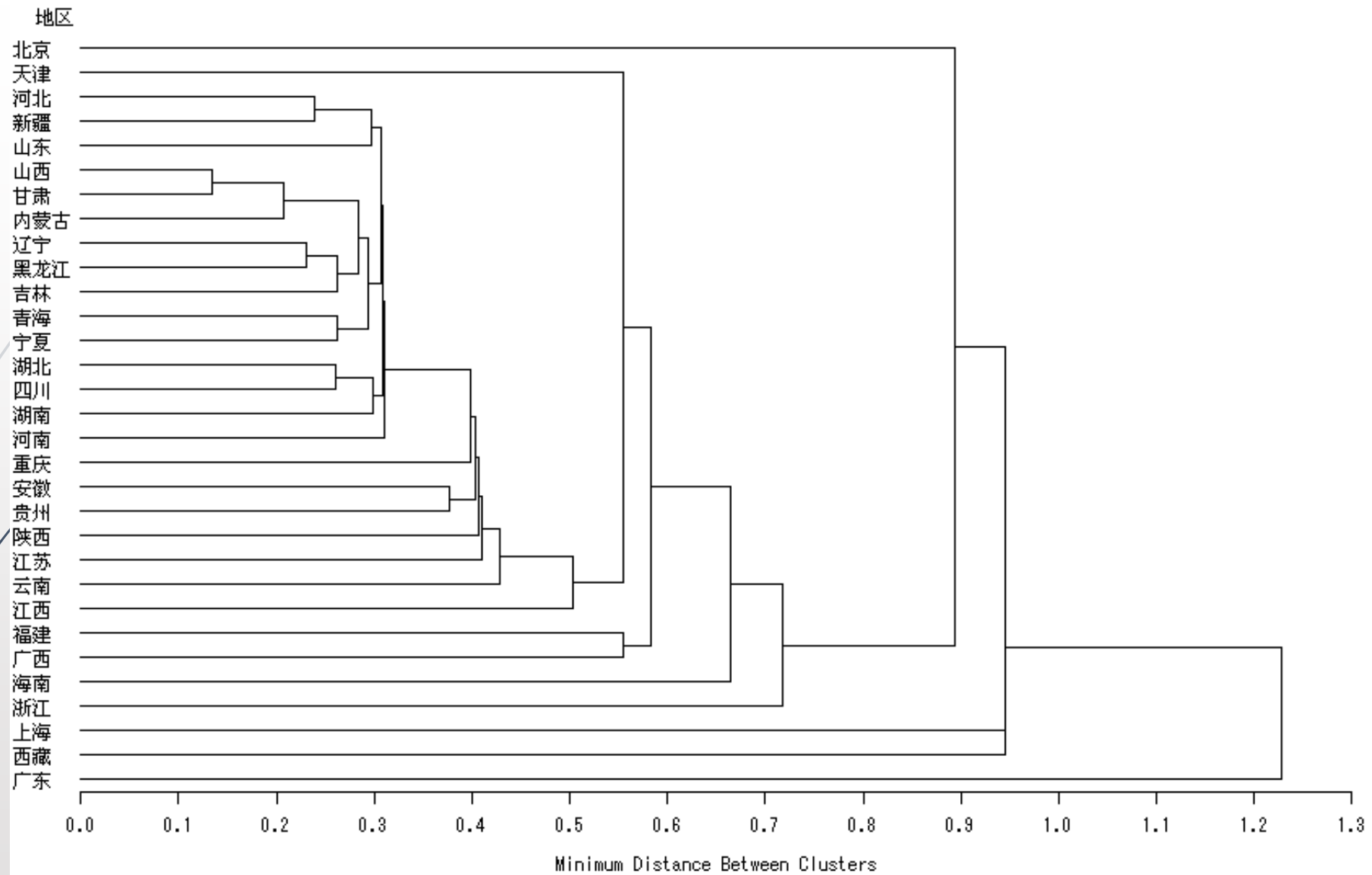


Figure 6.3.12 Shortest Distance Method

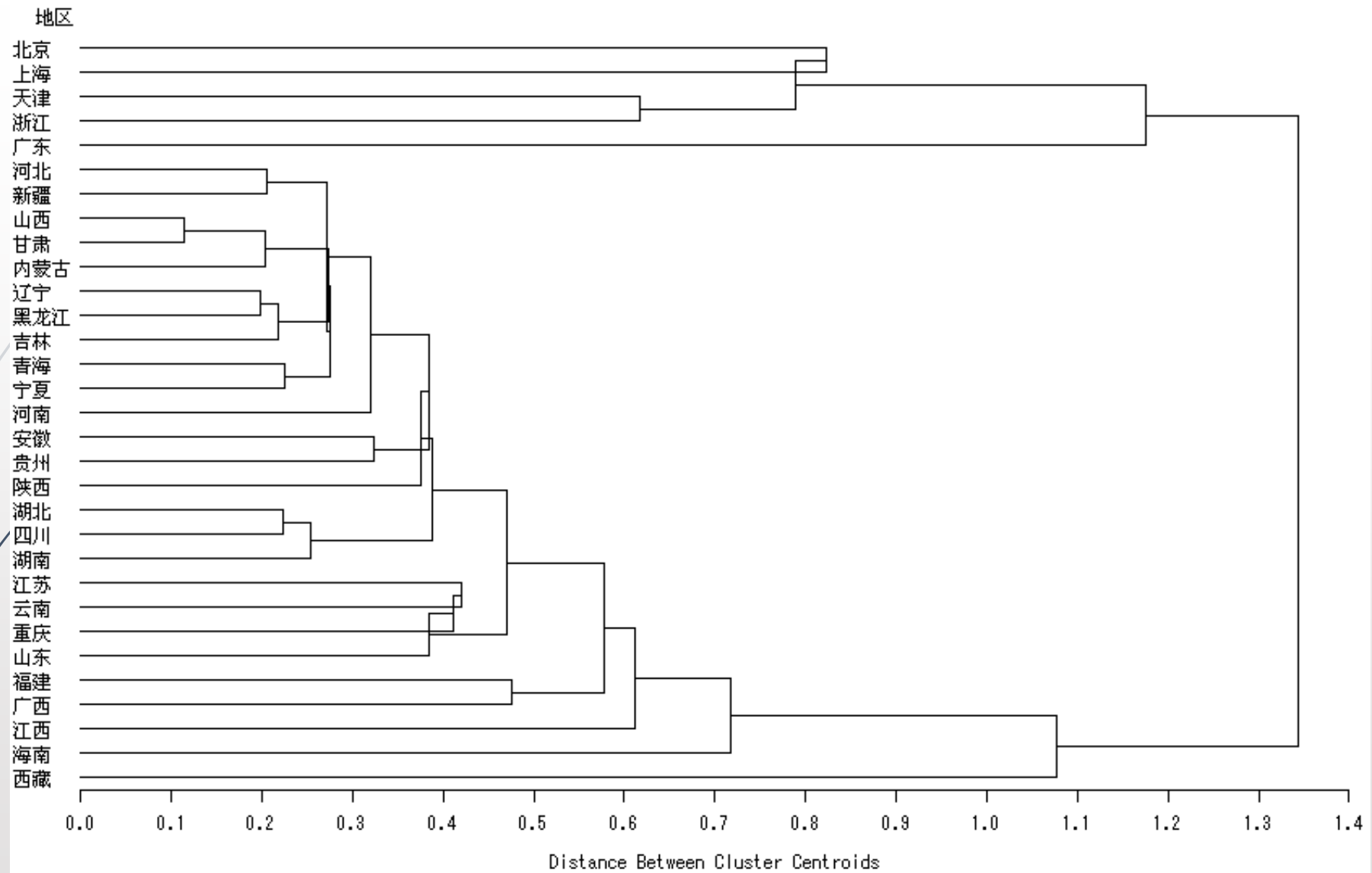


Figure 6.3.13 Centroid Method

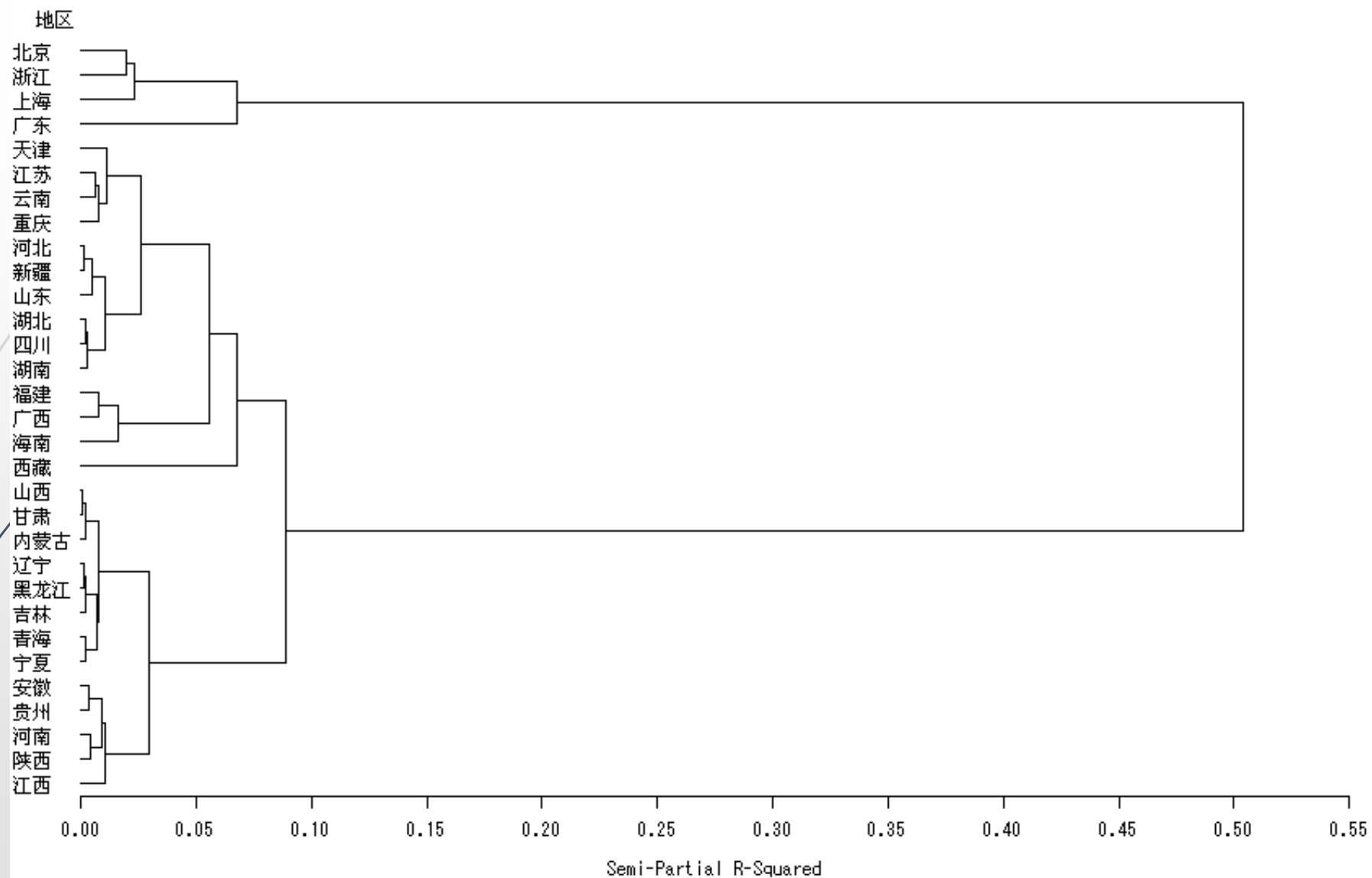


Figure 6.3.14 Sum of squares of deviations method

- According to the 3 tree diagrams, only Ward method' results conform to the real requirement for cluster. It divides 31 areas into the following 3 clusters :
 - I:** 北京、浙江、上海 and 广东. These are coastal areas with the most prosperous economy and the highest urban household consumption level.
 - II:** 天津、江苏、云南、重庆、河北、新疆、山东、湖北、四川、湖南、福建、广西、海南 and 西藏. These areas are in the intermediate level of economic development and urban household consumption.
 - III:** 山西、甘肃、内蒙古、辽宁、黑龙江、吉林、青海、宁夏、安徽、贵州、河南、陕西 and 江西. These areas are in the low level of economic development and urban household consumption.
- If they are divided into 5 clusters, then each of 广东 and 西藏 will be one cluster.

Monotony

- ▶ Let D_i denote the distance at the i —th iteration of hierarchical clustering. If a hierarchical clustering method satisfies $D_1 \leq D_2 \leq D_3 \leq \dots$, it is said to exhibit monotonicity. This monotonicity aligns with the concept of hierarchical clustering, where closer clusters are merged first before merging more distant ones.
- ▶ The shortest distance method, longest distance method, average linkage method, and sum of squares of deviations method exhibit monotonicity. However, the median method and centroid method do not possess this property.

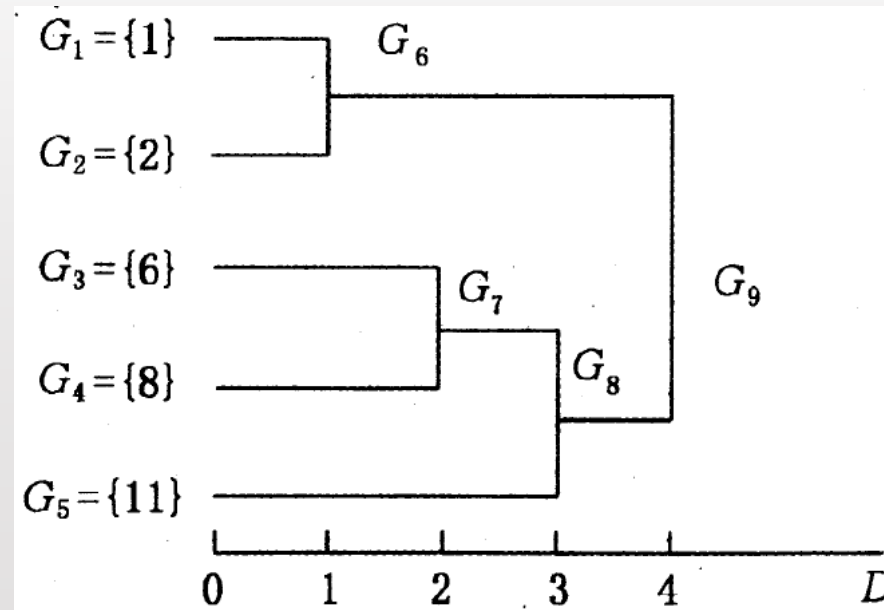
7. Numbers of clusters

- If samples can be distinctly divided into several well-separated clusters, determining the number of clusters is straightforward. Conversely, if the samples are difficult to separate clearly, determining the cluster number becomes challenging.
- Common methods of determining number of clusters :
 1. Give a threshold value T .
 2. Observe scatterplot of samples.
 3. Utilize statistic.

1. Give a threshold value T .

- By observing tree diagram, give a proper threshold value T to make all distances between clusters larger than T .

➤ Figure 6.3.2 Tree Diagram (Example 6.3.1)



$$T = 3.3$$

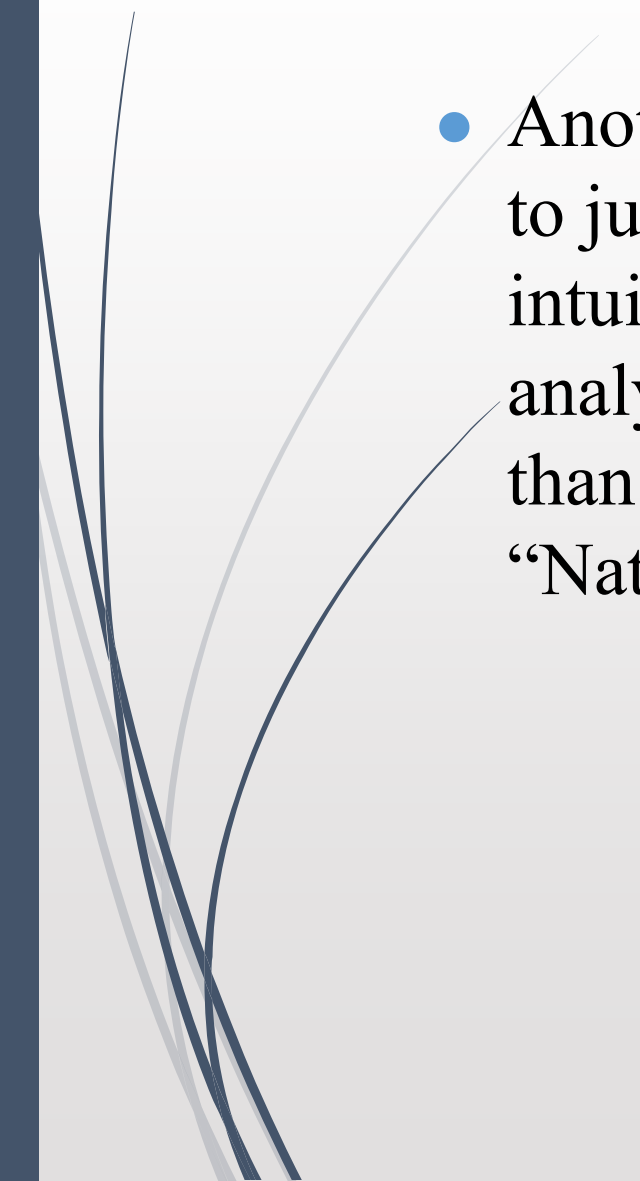
- Strong subjectivity is a drawback of this method.

2. Observe scatterplot of samples.

- If samples only have 2 or 3 variables, we can ensure cluster number by observing scatterplot. For 3 variables, we can spin 3D coordinate axis in SAS to observe scatterplot.
- If variables are more than 3, we can reduce dimensions to each considered cluster results through Fisher Discriminant Method. Draw a scatterplot by the former 2 or 3 discriminants of samples and observe the separate effect of clusters. This graph can not only help us evaluate the clustering effect, but also help us judge whether the number of classes is appropriate



Subjective Cluster from Scatterplot

- Another significant application of observing scatterplot is to judge the reasonability of the used cluster method intuitively. Sometimes we can do subjective cluster analysis directly from scatterplot. The effect may be better than formal cluster method, especially in searching for “Natural” clusters.
- 

Search for “Natural” Cluster



3. Utilize Statistics

- (1) R^2 Statistics

k : the number of clusters to be determined;

k clusters: $G_1, G_2, \dots, G_k$,

n_i : the sample size of cluster G_i ($i = 1, 2, \dots, k$);

Total sample size $n = \sum_{i=1}^k n_i$.

The overall centroid of all samples $\bar{x} = \frac{1}{n} \sum_{i=1}^n n_i \bar{x}_i$;

Sum of squared deviations for all samples:

$$W = \sum_{j=1}^{n_i} (x_j - \bar{x})' (x_j - \bar{x})$$

Sum of squared deviations for cluster G_i :

$$W_i = \sum_{j \in G_i} (x_j - \bar{x}_i)' (x_j - \bar{x}_i)$$

Sum of within-cluster sums of squared deviations across all K clusters:

$$P_k = \sum_{i=1}^k W_i$$

3. Utilize Statistics

- (1) R^2 Statistics

W can be decomposed as follows:

$$\begin{aligned} W &= \sum_{j=1}^n (x_j - \bar{x})' (x_j - \bar{x}) \\ &= \sum_{i=1}^k \sum_{j \in G_i} (x_j - \bar{x})' (x_j - \bar{x}) \\ &= \sum_{i=1}^k \sum_{j \in G_i} (x_j - \bar{x}_i + \bar{x}_i - \bar{x})' (x_j - \bar{x}_i + \bar{x}_i - \bar{x}) \\ &= \sum_{i=1}^k [\sum_{j \in G_i} (x_j - \bar{x}_i)' (x_j - \bar{x}_i) + n_i (\bar{x}_i - \bar{x})' (\bar{x}_i - \bar{x}) \\ &\quad + 2 \sum_{j \in G_i} (x_j - \bar{x}_i)' (\bar{x}_i - \bar{x})] \\ &= P_k + \sum_{i=1}^k n_i (\bar{x}_i - \bar{x})' (\bar{x}_i - \bar{x}). \end{aligned} \tag{6.3.21}$$

Define

$$R^2 = 1 - \frac{P_k}{W} = \sum_{i=1}^k n_i (\bar{x}_i - \bar{x})' (\bar{x}_i - \bar{x}) / W \tag{6.3.22}$$

- A smaller value of P_k/W (i.e., a larger R^2) indicates that the proportion of within-cluster sum of squared deviations relative to the total sum is smaller, which in turn suggests that the k clusters are more clearly separated.
- $0 < R^2 < 1$, it decreases as the number of clusters is reduced.

3. Utilize Statistics

- (1) R^2 Statistics

$$R^2 = 1 - \frac{P_k}{W_k} = \sum_{i=1}^k n_i (\bar{x}_i - \bar{x})' (\bar{x}_i - \bar{x}) / W \quad (6.3.22)$$

- A smaller value of P_k/W_k (i.e., a larger R^2) indicates that the proportion of within-cluster sum of squared deviations relative to the total sum is smaller, which suggests that the k clusters are more clearly separated.
- $0 < R^2 < 1$, it decreases as the number of clusters is reduced.
- $R^2 = 1$ if $k = n$; $R^2 = 0$ if $k = 1$.
- In general, we aim to keep k as small as possible while maintaining a relatively high R^2 value. Therefore, any further reduction in k should not come at the cost of a significant drop in R^2 .
- For example, suppose that during the merging process leading up to four clusters, the decrease in R^2 is gradual. When the data are grouped into four clusters, $R^2 = 0.8$, but after the next merge ($k = 3$), R^2 drops sharply to 0.32. In this case, $k = 4$ is thought to be the most appropriate.

3. Utilize Statistics

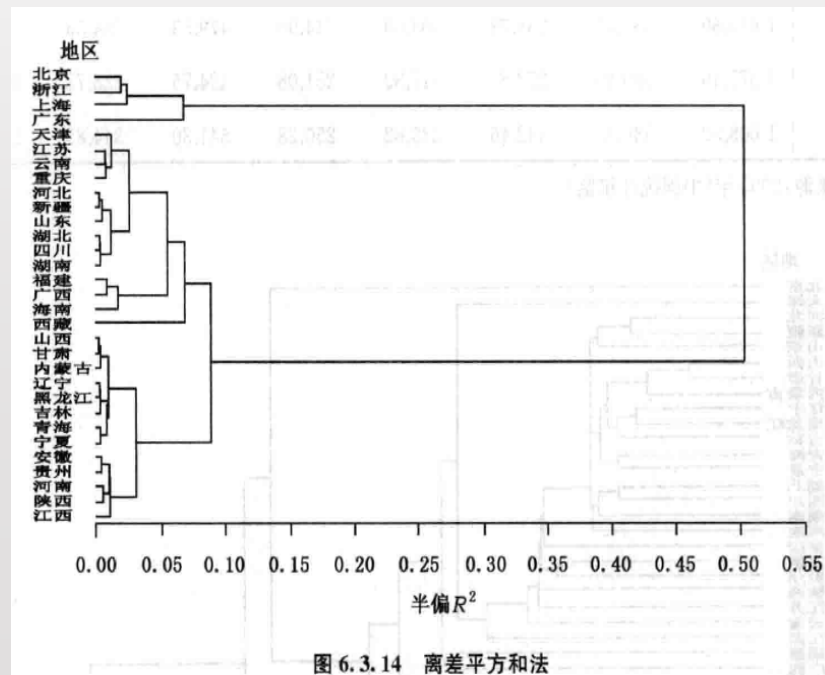
- (2) Semi-partial R^2 Statistics:

$$\tilde{R}^2 = \frac{D_{kl}^2}{W} = 1 - \frac{W_k - W_l}{W} - \left(1 - \frac{W_M}{W}\right) = R_{pre}^2 - R_{cur}^2,$$

where $D_{kl}^2 = W_M - W_k - W_l$ (并类类内离差平方和的增量).

- $\tilde{R}^2$ is defined as the difference between the R^2 value from the previous step and that of the current step. A larger $\tilde{R}^2$ indicates that the clustering result from the previous step was more effective in preserving the explained variance.
- Figure 6.3.14

Clustering the data into three groups is the most appropriate.



3. Utilize Statistics

- (3) Pseudo (伪) F-statistics

$$\text{Pseudo } F = \frac{(W - P_k)/(k-1)}{P_k/(n-k)} = \frac{n-k}{k-1} \frac{R^2}{1-R^2} \quad (6.3.24)$$

- $\frac{R^2}{1-R^2}$ has the same role as R^2 , it decreases as the number of clusters is reduced.
- $\frac{n-k}{k-1}$ can be viewed as an adjustment coefficient that helps prevent the pseudo- F value from decreasing simply due to a reduction in the number of clusters.
- This allows us to directly use the magnitude of the pseudo F value to determine the most appropriate number of clusters. A larger pseudo- F value indicates a better clustering result.

3. Utilize Statistics

(4) Pseudo t^2 -statistics

$$\begin{aligned} \text{Pseudo } t^2 &= \frac{D_{kl}^2}{(W_k + W_l)/(n_k + n_l - 2)} \\ &= (n_k + n_l - 2) \left(1 - \frac{R_{cur}^2}{R_{pre}^2} \right) \end{aligned} \quad (6.3.25)$$

- A larger pseudo t^2 value indicates that the two merged clusters were well separated prior to merging, meaning the previous clustering step was effective.
- The pseudo t^2 serves a similar purpose to that of the partial R^2

§ 6.4 Dynamic Clustering (动态聚类法)

- In the system clustering method, the samples that were previously "incorrectly" classified are no longer provided with the opportunity of reclassification, while the dynamic clustering method allows the samples to move from one class to another.
- **Dynamic Clustering**'s basic idea is to choose a batch of condensation points (凝聚点) or provide an initial clustering. Condensate samples to condensation points following a certain principle and continuously modify and iterate condensation points until we obtain reasonable clusters or stable iteration. **The number of clusters k need be specified in advance.** A simple method to select initial condensation points (or initial clusters) is data splitting(样本分割).

k means method's basic steps

Dynamic clustering includes many methods. In this section, we only discuss one popular dynamic clustering method——*k means method*. *K* means method is an algorithm introduced and named after MacQueen (1967) .

- ➡ (1) Choose k samples as initial condensation points or just divide them into k initial clusters and choose the center of gravity of k clusters as initial condensation points.
- ➡ (2) For all samples, each sample is classified into the class whose condensation point is nearest to it (usually measured by Euclidean distance). Condensation points are replaced by mean values of the current cluster until all samples are classified.
- ➡ (3) Repeat Step(2) until all samples are classified.

- To some extent, the final clustering result depends on the selection of initial condensation points or initial clusters. Most significant change in clustering process happened during first redistribution process.
- **Example 9.4.1** Cluster example 9.3.1 using k averaging method, suppose $k=2$. Specific steps:
 - (1) Randomly divide samples into two clusters $G_1^{(0)} = \{1, 6, 8\}$ and $G_2^{(0)} = \{2, 11\}$. Then mean values of initial clusters are 5 and $6\frac{1}{2}$.
 - (2) Calculate Euclidean distance between 1 and two clusters(mean values)

$$d(1, G_1^{(0)}) = |1 - 5| = 4$$

$$d(1, G_2^{(0)}) = \left| 1 - 6\frac{1}{2} \right| = 5\frac{1}{2}$$

Since distance between 1 and $G_1^{(0)}$ is shorter than that of $G_2^{(0)}$.

Therefore, 1 needn't redistribution. Calculate distances between 6 and two clusters:

$$d(6, G_1^{(0)}) = |6 - 5| = 1$$

$$d(6, G_2^{(0)}) = \left| 6 - 6\frac{1}{2} \right| = \frac{1}{2}$$

6 should be redistributed to $G_2^{(0)}$. After modification, two clusters are $G_1^{(1)} = \{1, 8\}$ and $G_2^{(1)} = \{2, 6, 11\}$ and mean values are $4\frac{1}{2}$ and $6\frac{1}{3}$.

Calculate

$$d(8, G_1^{(1)}) = \left| 8 - 4\frac{1}{2} \right| = 3\frac{1}{2}$$

$$d(8, G_2^{(1)}) = \left| 8 - 6\frac{1}{3} \right| = 1\frac{2}{3}$$

8 is reclassified to $G_2^{(1)}$ and clusters are $G_1^{(2)} = \{1\}$ and $G_2^{(2)} = \{2, 6, 8, 11\}$. Mean values are 1 and $6\frac{3}{4}$. Calculate

$$d(2, G_1^{(2)}) = |2 - 1| = 1$$

$$d(2, G_2^{(2)}) = \left| 2 - 6\frac{3}{4} \right| = 4\frac{3}{4}$$

Redistribute 2 to $G_1^{(2)}$, two clusters are $G_1^{(3)} = \{1, 2\}$ and $G_2^{(3)} = \{6, 8, 11\}$. Mean values $1\frac{1}{2}$ and $8\frac{1}{3}$.

- (3) Calculate the distance between each sample and mean value when

$$G_1^{(3)} = \{1, 2\} \text{ and } G_2^{(3)} = \{6, 8, 11\}.$$

Table 6.4.1 Distance between samples and mean values

类 \ 样品	1	2	6	8	11
$G_1^{(3)} = \{1, 2\}$	$\frac{1}{2}$	$\frac{1}{2}$	$4\frac{1}{2}$	$6\frac{1}{2}$	$9\frac{1}{2}$
$G_2^{(3)} = \{6, 8, 11\}$	$7\frac{1}{3}$	$6\frac{1}{3}$	$2\frac{1}{3}$	$\frac{1}{3}$	$2\frac{2}{3}$

- Final clustering results are $\{1, 2\}$ and $\{6, 8, 11\}$.

Example 6.4.2. Example 6.3.3 is clustered using the K-means method. All variables are standardized before clustering. The SAS codes for clustering are as follows:

```
/*clear all*/
DM ' log; clear; odsresult; clear; output;
clear; ';

libname sasuser "/home/u61383528";
/* sasuser为缩略的文件名, 代指/home/u61383528 */
/*import data*/
PROC IMPORT DATAFILE =
  '/home/u61383528/examp6.3.3.csv'
  OUT = sasuser.examp633;
RUN;

proc standard data=sasuser.examp633 mean=0 std=1
out=stan;
proc fastclus data=stan maxc=5 drift list;
  id region;
run;
```

聚类列表			
观测	region	聚类	与种子的距离
1	Beijing	1	1.8251
2	Tianjin	3	2.0554
3	Hebei	4	1.1024
4	Shanxi	4	0.7316
5	Neimenggu	4	0.9438
6	Liaoning	4	1.1938
7	Jilin	4	0.8209
8	Heilongjiang	4	1.1870
9	Shanghai	1	1.9257
10	Jiangsu	3	0.8092
11	Zhejiang	1	1.8002
12	Anhui	4	1.3410
13	Fujian	3	1.8337
14	Jiangxi	4	2.1449
15	Shandong	3	1.6212
16	Henan	4	0.9296
17	Hubei	4	1.4707
18	Hunan	3	0.7278
19	Guangdong	2	0
20	Guangxi	3	2.0164
21	Hainan	4	2.7708
22	Chongqing	3	1.3335
23	Sichuan	3	1.0456
24	Guizhou	4	1.2331
25	Yunnan	3	1.3094
26	Xizang	5	0
27	Shannxi	4	1.3131
28	Gansu	4	0.8220
29	Qinghai	4	1.1709
30	Ningxia	4	1.4135



Example 6.4.2. Example 6.3.3 is clustered using the K-means method. All variables are standardized before clustering. The clustering results are as follows:

- Cluster I: 北京、上海、浙江
- Cluster II: 广东
- Cluster III: 天津、江苏、福建、山东、湖南、广西、重庆、四川、云南
- Cluster IV: 河北、山西、内蒙古、辽宁、吉林、黑龙江、安徽、江西、河南、湖北、海南、贵州、陕西、甘肃、青海、宁夏和新疆。
- Cluster V: 西藏。



Assignments has been published on BB

